

You start with a directory of subdirectories which correspond to the recordings for a given day. If the top-level directory contains too many files, it may be worth organizing the subdirectories into smaller groups. `bats_transformer/copy_audio_files.py` can be used for this, as follows:

```
python3 copy_audio_files.py -i /qnab/bats/jasperridge/lake2/audio/ -o /qnab/bats/jasperridge/lake2/grouped_audio
```

Once you have the “directory of subdirectories of audio files” structure that you want, go to SonoBat Data Wizard, and select the “Long File Parser” action.

Use the following parameters (see picture below):

- The box at the middle-top (*Search for .wav files and subfolders of .wav files*) should be highlighted by default, and keep it that way.
- Change the “Maximum output file length” to “2 seconds”.
- Change the “Frequency band selector” to “Search 5 kHz and above for bats”
- Leave all other fields as is.

The screenshot shows the SonoBat Data Wizard window. The 'Long File Parser' tab is selected. The search path is set to `\\wsl.localhost\Ubuntu\home\ayc227\bats\audio\audio_07_22\original\20220701-04`, and it shows 1313 files found. The 'Output Folder' is empty. The 'Filename Prefix' is empty. The 'Acceptance level' is set to '(tolerant) accepts files with shorter, less strong calls, but more noise'. The 'Maximum output file length' is set to '2 seconds'. The 'Frequency band selector' is set to 'Search 5 kHz and above for bats'. The 'Filter selection' is set to 'autofilter'. The 'Channel selection if stereo' is set to 'If Stereo, scan right channel'. A 'Process' button is at the bottom right.

SonoBat Data Wizard

Search for .wav files and subfolders of .wav files 1313 files found

Long File Parser

`\\wsl.localhost\Ubuntu\home\ayc227\bats\audio\audio_07_22\original\20220701-04` exit

Use the **Long File Parser** to detect and write to separate files the individual bat pass events within long duration recordings, for example like those produced by devices such as AudioMoth detectors. Even with some devices that trigger recordings using basic amplitude triggering you may still prefer to record continuous files and use the more sophisticated full-spectrum based call and sequence recognition of this utility to parse recording data into logical bat pass sequences to produce more accurate bat activity counts. *Note: date and time stamps of parsed files reckon from finding a date and time stamp in the filename of the long file having the format YYYYMMDD_hhmmss. Note 2: you must have permission to write to the directories involved.*

Output Folder:

If left blank, parsed files will write to a folder named " 'Filename Prefix of the Long File'_Parsed Files" at the same directory level as the long file. If the long file has not filename prefix then parsed file will write to a folder name " 'Parent Folder Name'_Parsed Files" at the same level as the long file. This provides a better option for file management if processing a batch of files.

Filename Prefix for parsed files (parsed files will have the filename format " 'Filename Prefix'-YYYYMMDD_hhmmss.wav"):

If left blank, parsed files will inherit any filename prefix from the parent long file. This provides a better option for file management if processing a batch of files.

Acceptance level: (tolerant) accepts files with shorter, less strong calls, but more noise

Maximum output file length: 2 seconds

The Long File Parser will save shorter sequences of calls as shorter duration files, but limit file length to this setting.

☐ Force file lengths to above selection regardless of content?

Frequency band selector Search 5 kHz and above for bats

Filter selection: autofilter

Channel selection if stereo: If Stereo, scan right channel

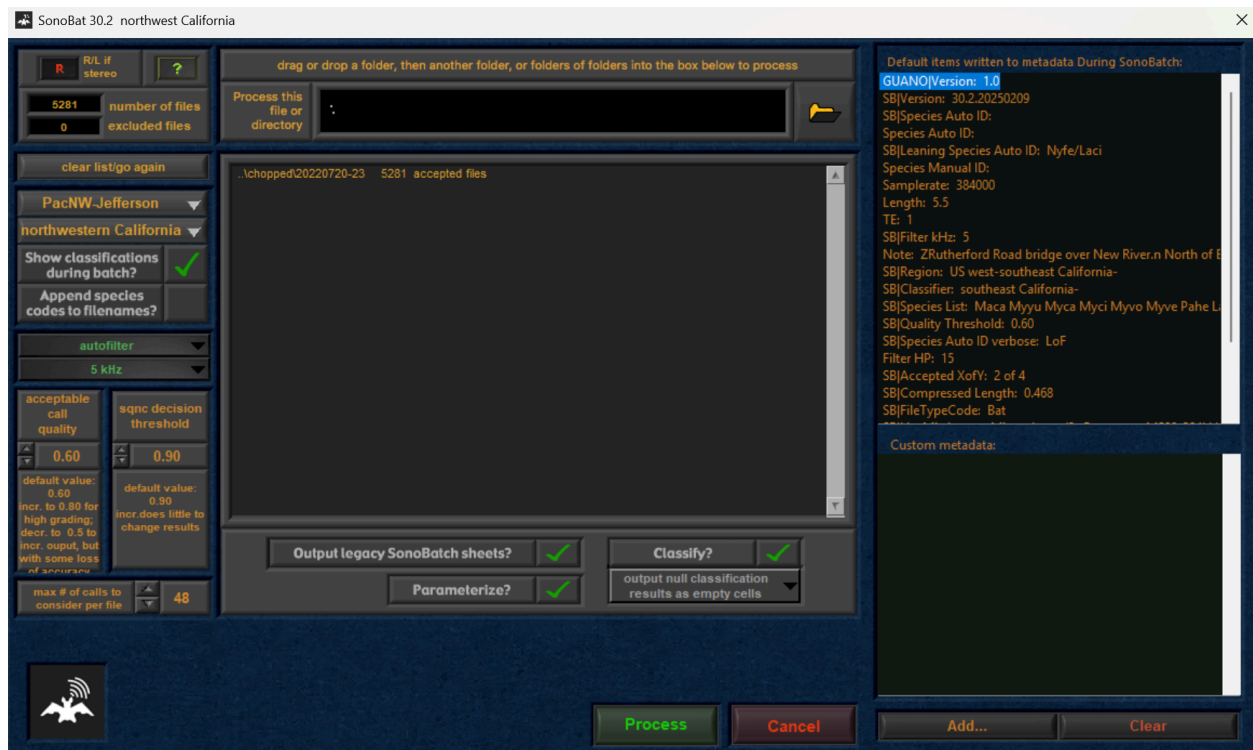
Process

This will put a new directory in each of the subdirectories (by day) that contains the new shortened files. Gather these new directories and place them in a separate directory from the original files (this is unfortunately currently a manual bottleneck).

Take the new directories to SonoBatch (through the main SonoBat North America app, using the Northwest California setting in the PacNW pack).

Use the following settings:

- In “Process this file or directory”, insert your directory of subdirectories of parsed audio files
- Change “max # of calls to consider per file” on the bottom left to 48.
- Check “Output legacy SonoBatch sheets?” and subsequently “Parameterize?” and “Classify?”
- Keep the rest of the fields as default.



This will get all the chirp parameters and classify the audio files (though seemingly not the individual chirps). With these, we can easily generate the .feather datasets for model training.